

Executive Summary & System Architecture

Urban flood nowcasting in India faces an operational disconnect: IMD delivers macro-scale meteorological rainfall predictions (e.g. "60 mm across Central Delhi"), yet municipal authorities cannot determine which underpass or intersection will submerge. **AGASTYA** (Adaptive Geospatial Analysis & Stormwater Topology Yield Architecture) resolves this by dynamically coupling live precipitation feeds with high-resolution digital terrain slope (CartoDEM 10m) and synthetic underground drainage conduits generated via **PySewer**.

Rainfall Feed (Open-Meteo) → CartoDEM Slope → PySewer Drainage Graph → 1D Manning Hydraulics → Surge Depth (cm) → Constrained Dijkstra Routing

Core Benchmark: 25 topological nodes · 39 drainage conduits · <4.5ms hydraulic compute time · <2ms emergency routing · 53/53 automated tests passing · Full client-side offline dual-solver parity.

1. Empirical vs Synthetic Data Segregation

AGASTYA strictly segregates empirical real-world observational inputs from synthetic mathematical proxies, ensuring scientific defensibility:

Data Layer	Data Source & Resolution	Classification	Functional Role in AGASTYA
Precipitation Input	Open-Meteo API / IMD Radar (0.25° grid)	Empirical Real	Provides real-time hourly rainfall intensity (0–100 mm/hr) and storm duration curves without API keys.
Topography (DEM)	ISRO CartoDEM 10m / SRTM 30m	Empirical Real	Defines terrain slope and watershed boundary; reveals Minto Bridge underpass sag (210.5m) vs CP ridge (216.5m).
Road Network	OpenStreetMap / OSMnx graph extraction	Empirical Real	Supplies road segment lengths, coordinates, junction connectivity, and emergency traversal corridors.
Drainage Conduits	PySewer Synthetic Topology (JOSS 2024)	Synthetic Proxy	Overcomes India's missing pipe data deficit by synthesizing CPHEEO-compliant gravity pipes (300–1800mm, n=0.015).

2. Mathematical Formulation of Hydraulic Engine

Traditional 2D hydrodynamic tools (e.g. EPA-SWMM) require 30–45 minutes per iteration, making real-time nowcasting impossible. AGASTYA introduces a lightweight 1D open-channel pipe conveyance solver coupled with rational surface overland surcharge on a topological graph:

2.1 Surface Runoff Inflow (Rational Method)

$Q_{in,i} = (C_i \cdot I \cdot A_i) / 360$ [m³/s] Where: $C_i = 0.85$ (dense urban asphalt/concrete), I = rainfall rate (mm/hr), A_i = sub-catchment area (hectares)

2.2 Conduit Gravity Flow Capacity (Manning's Equation)

$Q_{cap,ij} = (1 / n) \cdot A_{pipe} \cdot R^{(2/3)} \cdot S^{(1/2)}$ [m³/s] Where: $n = 0.015$ (roughness), $A_{pipe} = \pi \cdot D^2 / 4$, $R = D / 4$ (circular pipe flowing full), $S = |Elev_i - Elev_j| / Length_{ij}$

2.3 Overland Inundation & Sag Ponding Concentration

$Q_{surcharge,i} = \max(0, Q_{in,i} - \sum Q_{cap,out} + Q_{backwater,in})$ Water_Depth_i (cm) = $(Q_{surcharge,i} \cdot \Delta t \cdot Duration_Scale / Ponding_Area_i) \cdot 100$ At Minto Bridge Sag (node_minto_bridge_sag, 210.5m): Downpour @ 75 mm/hr → Water Depth = 38.3 cm [CRITICAL RISK - Vehicle Impassable] Dry Baseline @ 0 mm/hr → Water Depth = 0.0 cm [SAFE - Dry Road]

3. Constrained Dijkstra Emergency Routing Architecture

Standard GPS mapping applications optimize purely for geometric transit distance, routinely routing emergency ambulances into flooded underpass depressions. AGASTYA integrates dynamic vehicular depth gating:

Vehicle Inundation Safety Threshold (15.0 cm): Standard ambulance chassis (e.g. Force Traveller, Tata Winger) operate with engine intake and tailpipe clearances of 200–250 mm. Traversal through water depths exceeding 15.0 cm risks mechanical hydrolock, electrical stall, and patient demise during the golden hour.

3.1 Strict Endpoint Safety Lifecycle

Prior to invoking Dijkstra graph relaxation, AGASTYA enforces strict endpoint safety gates:

Routing Condition	Hydraulic Depth Check	System Safety Action	Operational Dispatch Outcome
Safe Dry Corridor	Origin ≤ 15cm & Dest ≤ 15cm & all path edges ≤ 15cm	ROUTE_COMPUTED	Displays green arterial route overlay with distance and turn-by-turn ETA.
Flooded Sag Bypass	Origin & Dest safe; Minto Bridge underpass sag = 38.3 cm	DYNAMIC_REROUTE	Prunes flooded sag edges; calculates dry alternate detour via DDU Marg East (+2.1 mins).
Flooded Origin Node	Origin node water depth > 15.0 cm	ORIGIN_UNSAFE	Refuses unsafe deployment; prompts dispatch to mobilize alternate ambulance depot.
Flooded Destination	Destination node water depth > 15.0 cm	DESTINATION_UNSAFE	Refuses unsafe delivery; directs ambulance CAD to non-submerged trauma facility.
Completely Isolated Node	All egress road segments > 15.0 cm	NO_SAFE_ROUTE	Triggers municipal NDRF / flood rescue boat deployment request.

4. 53-Test Automated Regression Suite Results

The AGASTYA backend and hydraulic logic is validated by 53 automated unit and integration tests passing at 100%:

Test Subsystem	Tests	Verification Assertions	Result
Baseline Hydraulic Physics	12	Zero rain yields strictly 0.0 cm depth; Manning capacity monotonically rises with pipe diameter; Rational runoff scales linearly with rainfall rate.	12 / 12 PASS
Sag Basin Concentration	10	Lowest elevation (210.5m) concentrates overland surcharge (38.3 cm @ 75 mm/hr); higher elevations (216.5m) drain safely (<2.0 cm).	10 / 10 PASS
Multi-Node Choke Engine	11	Simultaneous manhole blockage; upstream backwater propagation; unblock state restoration; zero side-effects when choke switch is off.	11 / 11 PASS
Endpoint Safety & Edge Pruning	12	Submerged origin returns ORIGIN_UNSAFE; submerged destination returns DESTINATION_UNSAFE; flooded edges (>15 cm) pruned from graph.	12 / 12 PASS
Offline Client Parity & Schema	8	Full client-side TypeScript physical and Dijkstra solver mirrors Python backend; Pydantic v2 schemas pass serialization.	8 / 8 PASS

5. Visual Verification Evidence: Test Cases A & B



Figure 1 (Test Case A — Submerged Origin Endpoint): Rainfall set to 75 mm/hr downpour; Origin node water depth = 17.5 cm (>15.0 cm threshold). System immediately halts dispatch with amber alert: ORIGIN_UNSAFE: Origin node is submerged (depth > 15cm). Ambulance cannot deploy. No unsafe driving route is rendered on screen.



Figure 2 (Test Case B — Submerged Destination Endpoint): Destination node water depth = 19.8 cm (>15.0 cm threshold). System triggers red alert banner: DESTINATION_UNSAFE: Destination node is submerged (depth > 15cm). Rerouting to alternate medical facility. Prevents emergency ambulance from being stranded at a flooded trauma facility entrance.

6. Visual Verification Evidence: Test Case C (Flooded Sag Bypass)



Figure 3 (Test Case C — Flooded Sag Bypass): Minto Bridge underpass sag is inundated to 38.3 cm (CRITICAL, highlighted in red). AGASTYA automatically prunes the inundated underpass corridor and computes an active green arterial bypass via DDU Marg East in <2 milliseconds, saving 11 minutes of transit and preventing vehicle drowning.

7. Competitive Advantage vs Existing Solutions

Key Capability	Traditional Flood Portals	Commercial Flood APIs	Municipal ICCC Dashboards	AGASTYA (Our Solution)
Street-Level Depth (cm)	× (City-wide only)	▲ (Satellite flood extent)	× (Qualitative text)	✓ (Exact cm per node)
Drainage Pipe Coupling	× (Ignored)	× (Overland only)	× (Static asset list)	✓ (Manning 1D physics)
Missing Pipe Synthesis	× (Requires CAD maps)	× (Requires sensors)	× (Missing data barrier)	✓ (PySewer AI synthesis)
Safe Ambulance Routing	× (No routing)	× (Historical maps)	▲ (Manual radio dispatch)	✓ (Dijkstra <15cm gate)
Submerged Endpoint Refusal	× (No safety checks)	× (No vehicle logic)	× (Unsafe dispatch)	✓ (ORIGIN / DEST UNSAFE)
Multi-Node Choke Simulation	× (Static models)	× (No clog modeling)	× (Reactive complaints)	✓ (Interactive toggle)
100% Offline Resilience	× (Crashes offline)	× (Cloud-dependent)	× (Server-dependent)	✓ (Client-side engine)
Deployment Cost	High (₹1–2 Cr consulting)	High (Recurring SaaS \$)	Extreme (₹5–20 Cr SCADA)	✓ (<₹10 Lakhs/city)

8. Supporting Facts for Feasibility & Civic Viability

Empirical Metric / Fact	Official Value	Authoritative Source	Operational Impact
Annual Flood Economic Loss	\$14 Billion (₹14,000+ Crore)	World Bank & NDMA Guidelines	Validates high civic ROI; preventing arterial inundation protects trade.
Public Utility Damage (2021)	₹25,244 Crore in a single year	Ministry of Home Affairs Disaster Stats	Enables Smart City ICCCs to safeguard electrical substations.
Urban Stormwater Deficit	70%+ urban India lacks drain maps	CPHEEO / MoHUA Urban Reports	Demonstrates critical need for synthetic drainage generation (PySewer).
Flood Mortality (2024)	361 deaths (2,754 in 2019)	NCRB Accidental Deaths Report (ADSI)	Saves human lives by preventing vehicles from drowning in underpasses.
Addressable Market Size	4,000+ Urban Local Bodies (ULBs)	Smart Cities Mission, Govt of India	₹2,000+ Crore addressable market at ₹5–10L/year per municipal licence.

9. Scientific Research Citations & Policy References

Domain / Research Topic	Official Reference & URL	Key Technical Contribution
Precipitation Nowcasting	IMD Gridded Dataset (cdsp.imdpune.gov.in)	0.25° × 0.25° historical precipitation records and radar coverage standards.
Digital Elevation Modeling	ISRO CartoDEM / Bhuvan (bhuvan.nrsc.gov.in)	10-meter digital elevation model for watershed boundaries and terrain slope.
Synthetic Sewer Generation	PySewer (JOSS 2024, doi:10.21105/joss.05834)	Peer-reviewed methodology for synthesizing gravity drainage networks.
Urban Drainage Standards	CPHEEO Stormwater Drainage Manual (MoHUA)	Indian design criteria for stormwater conveyance and Manning roughness (n=0.015).
Street Graph Extraction	OSMnx / OpenStreetMap (osmnx.readthedocs.io)	Complex network extraction for road graph topology and navigation traversal.
Disaster Risk Governance	NDMA Urban Flood Guidelines	National policy framework for urban flood early warning and hospital evacuation.

10. 6-Member Team Contribution Matrix

Team Member	Assigned Role	Core Technical Responsibilities	Delivered Artifacts
Deepanshu (Lead)	Lead System Architect	End-to-end architecture, FastAPI backend, pitch delivery, technical defense.	Core platform, technical report, pitch script.
Member 2	Hydrologic / GIS Engineer	CartoDEM slope sampling, PySewer pipe synthesis, Manning capacity modeling.	25-node topology graph, elevation model.
Member 3	Frontend UI/UX Specialist	React 19 dashboard, Leaflet mapping visualizer, color tokens, interactive controls.	Interactive operator UI, glassmorphic HUD.
Member 4	Algorithm & Routing Lead	Dijkstra emergency solver, vehicle water clearance gates, edge pruning.	Routing engine, endpoint safety logic.
Member 5	QA & Reliability Engineer	Automated pytest suite (53 tests), regression auditing, browser verification.	Passing test suite, regression matrix, demo videos.
Member 6	Policy & Research Analyst	Economic loss data collection, NDMA/CPHEEO compliance, market benchmarking.	Facts sheet, comparison matrix, references.

11. 36-Hour Hackathon Implementation Roadmap & Milestones

Milestone Phase	Build Timeline	Technical Objectives Completed	Verified Deliverable
Phase 1: Ingestion	Hours 00–06 · Geospatial	OSMnx road graph extraction for Minto basin; CartoDEM 10m elevation interpolation; Open-Meteo API pipeline.	Topological road graph & DEM raster.
Phase 2: Hydrology	Hours 06–16 · PySewer	Synthesized 25-node, 39-conduit pipe network; calibrated CPHEEO intake coefficients; verified 1D Manning flow.	Hydraulic surcharge engine (<4.5ms).
Phase 3: Algorithms	Hours 16–24 · Safe Route	Implemented vehicular Dijkstra routing with 15.0 cm clearance gate; built ORIGIN / DESTINATION UNSAFE safeguards.	Emergency dispatch routing API (<2ms).
Phase 4: Interface	Hours 24–30 · Frontend HUD	React 19 + Leaflet map visualizer; interactive multi-node choke toggles; color token safety separation.	Live interactive operator console.
Phase 5: Audit	Hours 30–36 · Hardening	53/53 automated pytest suite; dual-solver client-side offline parity; cloud deployment on Render/Vercel.	100% verified live production system.

Production Readiness & Municipal Handover: AGASTYA is containerized with Docker, deployable via GitHub Actions CI/CD, and requires zero proprietary CAD licenses. Municipal corporations (NDMC, BMC, BBMP) can onboard any urban basin within 4 hours simply by configuring city bounding coordinates.